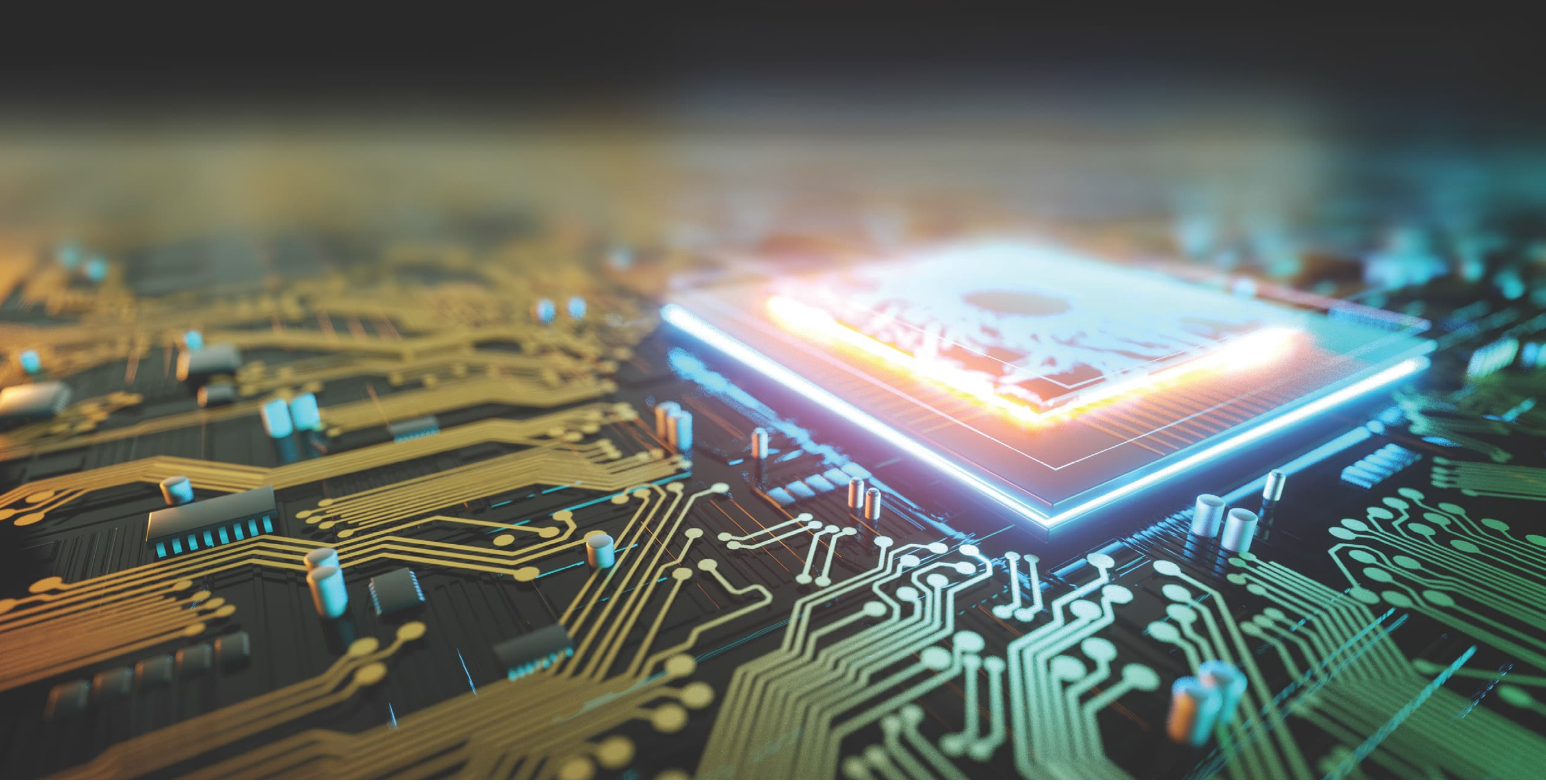


DIGITAL INDUSTRIES SOFTWARE

Shifting left with DFT to optimize productivity, testability and time-to-market

Executive summary

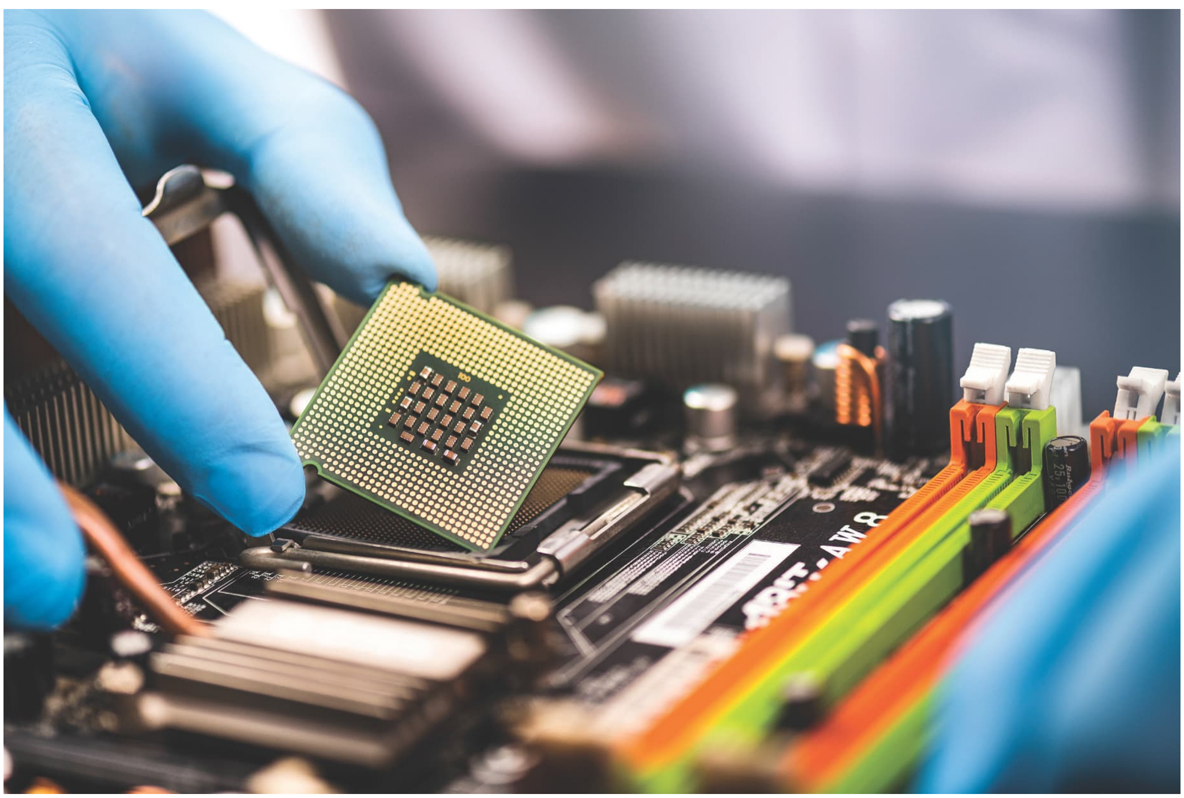
This paper discusses one of the Siemens EDA shift-left strategies in the RTL-to-signoff flow: shift-left design-for-test (DFT). Tessent™ RTL Pro software automates the analysis and insertion of Tessent VersaPoint™ test point technology, LBIST-OST test points, dedicated scan wrapper cells and x-bounding logic as behavioral code at the RTL level. Tessent RTL Pro builds on Tessent's market-leading DFT tools by providing unique functionality that helps customers shorten design turnaround time and accelerate time-to-market.

Jeff Mayer

Introduction

Creating competitive integrated circuit (IC) designs is a complex process involving multiple stages. The semiconductor industry constantly faces new challenges as design sizes grow, process nodes shrink, and engineering resources are increasingly stretched thin. These IC projects are expected to be high quality and reliable, yield well, and perform as intended. Design teams are tasked with keeping projects on schedule and within budget. To do this requires a high level of productivity throughout the design flow.

A shift-left strategy promises to deliver this improved productivity and product quality. Shifting left is the transference of some parts of the design flow to earlier stages. It involves multiple elements that impact the organization and its chances of success. Adopting a shift-left strategy allows engineers to learn more about their design earlier. This reduces iterations and the demand on engineering resources and improves overall design quality.



Why shift left?

Between advanced design pressures and limited engineering resources, IC companies need to streamline and optimize their design and implementation flows. Engineering costs are projected to increase nearly tenfold for direct costs and engineering time needed for all design stages. Shifting left for DFT-related tasks allows engineers to learn more about and fix their design problems earlier. This helps to reduce the impact of DFT on the critical path of chip design.

As the size of IC designs increases, impacts to product yield also become more visible, as a single silicon defect has a larger impact on the ability to deliver products to the end user. This situation is pushing the industry towards the heterogeneous integration of multiple dies within a single package. In this context of increasing design sizes and multi-die integration, there is also a rapidly growing demand for high-quality ICs for mission-critical applications, such as those in the automotive and aerospace sectors, which require ever shorter time-to-market.

One consequence of these trends is a significant increase in the total effort required for design and IC manufacturing. This has led to a strain on computing resources and a shortage of engineers, with many being stretched thin or unavailable. The semiconductor industry is tasked with meeting current market demands by producing IC products that achieve power, performance, area (PPA) and testability goals. The aim is to minimize design effort while maximizing cost-effectiveness, yield and resource utilization. Each stage of the design flow offers unique shift-left opportunities to support this goal.

IC designers use Tessent software to enhance the quality and reliability of their devices and to improve the efficiency of their DFT flow. A key trend in IC design is the implementation of a “shift-left” strategy. This approach involves addressing issues earlier in the RTL-to-signoff process by using data to solve problems sooner, effectively shifting the problem-solving phase to the left on the timeline.

Shift-left in DFT

A shift-left strategy brings new tools, techniques, and functionality to design teams. These help improve processes and design flows – whether to complete the same tasks more efficiently or to add new capabilities that solve challenges which arise during the design process. The practice of shifting left with DFT needs to include design analysis, test logic insertion, and verification of the DFT structures inserted in the design.

In legacy design flows, test insertion relied on a gate-level design and followed design synthesis (Figure 1a). Basic DFT components such as Memory BIST, scan and compression logic were included. This created an additional synthesis step after test logic insertion. The DFT logic RTL also required synthesis and whether this was done as part of a separate synthesis step or during design optimization, synthesis was typically run twice.

Tessent software can read the design RTL and insert DFT logic structures directly into the design. Shifting left with Tessent allows designers to perform synthesis and optimization of all design and DFT structures together. The DFT logic is inserted as user-readable behavioral code directly into design RTL. Figure 1b is an example of this modern design flow. It shows how this applies to all the main DFT logic components, such as Embedded Deterministic Test (EDT), Logic Built-In Self-Test (LBIST), Memory Built-In Self-Test (MBIST), On-chip Clock Control (OCC), Internal JTAG (IJTAG) logic, Streaming Scan Network (SSN), and others.

With the introduction of Tessent RTL Pro software, shift-left capabilities are expanded to include the remaining test logic that can be inserted at RTL. Tessent RTL Pro allows engineers to analyze their circuits and insert testability-related structures such as core wrapper cells, x-bounding logic and test points directly into the RTL. Tessent software also includes quick synthesis, enabling translation of behavioral RTL to basic cells. This type of basic synthesis enables scan insertion and Automatic Test Pattern Generation (ATPG) following RTL-based test logic insertion. Using this methodology, design teams can identify coverage and pattern count outliers in a fraction of the time, compared with running full synthesis (figure 2).

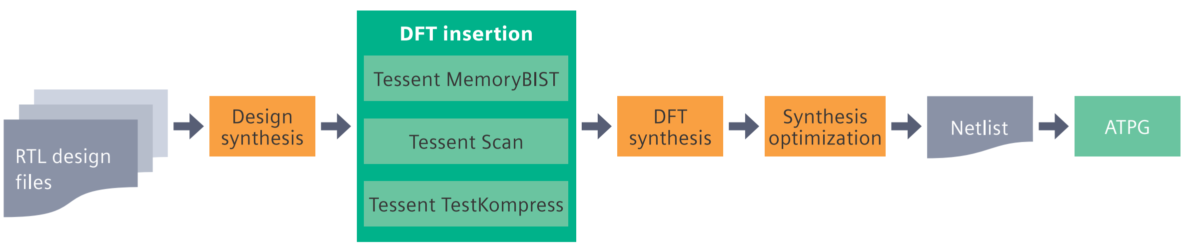


Figure 1a. Example of a legacy design flow, showing multiple time-consuming synthesis steps.

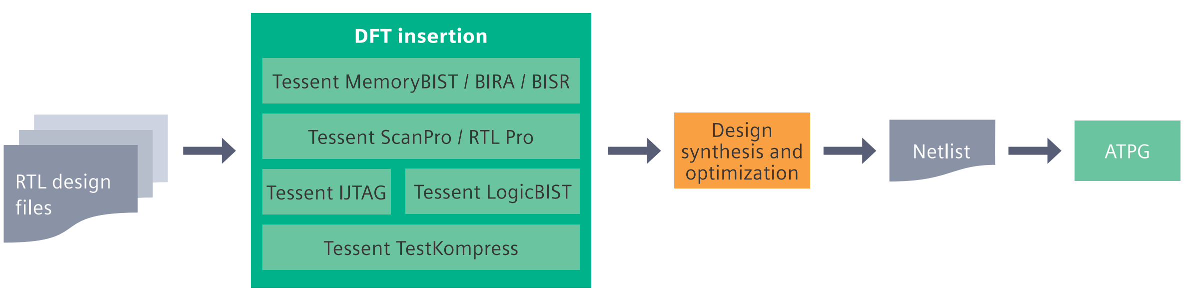


Figure 1b. Example of a modern design, showing DFT logic inserted directly into the design RTL and a single synthesis step which would include logic optimization.

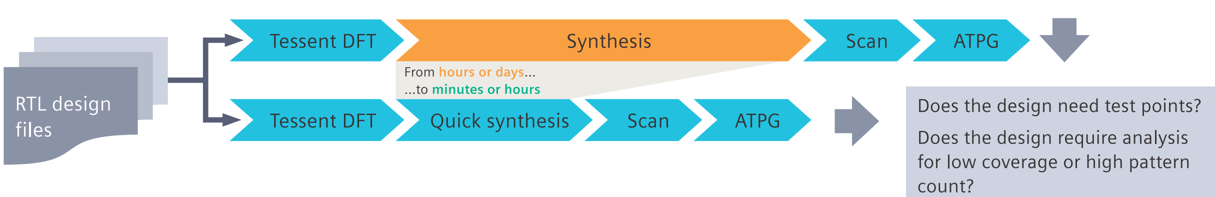


Figure 2. Quick synthesis shortens the time to learning and allows the earlier identification of test outliers.
Engineers can work on debugging test coverage and pattern count sooner rather than waiting for synthesis/optimization to complete.

Pre-synthesis and pre-DFT analysis

There are benefits to inserting test logic into the RTL:

- synthesis tools are able to perform optimization in the presence of DFT logic. This allows them to account for the impact of test logic on the timing and layout of the design; and
- designers can perform DFT design rule checks (DRCs) earlier in the design flow. Validation of clock, set and reset controllability for scan can be shifted left to improve productivity.

Tessent performs robust design rules checking depending on the DFT logic chosen for insertion. This allows the software to intelligently analyze the design, focusing only on the design paths requiring sensitization specifically for the DFT logic being inserted. These DRCs ensure that critical signals reach their intended destination in the appropriate test mode. Downstream, Tessent software analyzes the design and identifies inserted Tessent test logic, improving productivity and saving the designer time from manually identifying those inserted logic structures.

In expanding Tessent's shift-left flow enablement, designers can now insert X bounding, wrapper cells and test points directly into the RTL. The RTL, which is output from the Tessent software, maintains the same look and feel as the input RTL. This makes it easy to identify what was modified in the design files through visual inspection. Once all the test logic is inserted at RTL (except for scan), synthesis can better optimize for PPA and testability using the entire design in its near-complete form.

RTL wrapper and X-bounding analysis and insertion

Tessent performs wrapper and x-bounding analysis and insertion in the same analysis and insertion step as other logic components such as EDT, SSN, LBIST and OCC. After analysis, Tessent inserts dedicated wrapper cells for ports as needed.

To ensure that the synthesis process does not remove any essential wrapper cell logic, Tessent Scan has been enhanced with additional gate-level DRCs. These checks are performed to confirm that the test-related logic remains intact in its designated locations after synthesis optimization and continues to cover all necessary branches of the input and output ports. Examples of new DRCs include:

- Identify missing and pre-existing DWCs
- User-specified new DWC on a port with a pre-existing DWC
- Identified DWC not associated with any port
- Identified DWC doesn't cover all branches of wrapper I/O
- DWC specification doesn't match the design
- Unknown/unspecified DWC in design

Tessent adds bounding logic to block Xs, preventing the propagation of unknown values. Blocking unknown circuit values is required for logic BIST designs and helpful for ATPG designs. For logic BIST circuits, unknown states must be prevented from propagating to the Multi-Input Signature Register (MISR). For ATPG designs, blocking unknown states helps improve coverage and reduce pattern count, making it easier for pattern generation to target faults and merge test cubes.

Designers have the flexibility to control the Xs to be blocked. False and multi-cycle paths may be defined, allowing Tessent to insert logic, blocking the unknown states from reaching downstream observation points in the design. Both wrapper analysis and x-bounding at RTL include new pre-DFT DRCs to ensure that required logic may be inserted at the necessary locations in the RTL.

Test point analysis and insertion

Test point analysis and insertion at RTL is performed in a separate step, allowing for the application of scan-related design rules that ensure effective identification of test point locations. Test point analysis plays an important role in reducing pattern count for designs tested with ATPG. Tessent offers two types of specialty test points – VersaPoint test points and LBIST-OST (observation scan technology) [1, 2, 3] test points.

Tessent test points have demonstrated their effectiveness in ICs for automotive applications, which require in-system self-tests to be completed within a specific time frame to meet ASIL certification requirements. LBIST is utilized in these ICs as a safety monitor and is integrated into the functional logic. By inserting the test points for LBIST in the RTL design, there are fewer iterations needed to achieve sign-off, ensuring that safety considerations are incorporated into the RTL design from the outset rather than being treated as an afterthought.

Test points have proven their value in applications extending to other industries, including networking and data centers, where test time and cost significantly impact device profitability. VersaPoint test points, designed to reduce pattern count for ATPG designs, can decrease the pattern count by 2X to 5X compared to baseline coverage figures. (See "A new framework for RTL test points insertion facilitating a 'shift-left DFT' strategy," ITC NA 2023.)

```
always  @(posedge clk or negedge rst_n)
  begin
    if ((~rst_n))
      slot_cnt[7] <= 3'd0;
    else if (block_en_n)
      slot_cnt[7] <= 3'd0;
    else
    begin
      if ((((conf_ofd_mode & active_s[7]) & o_fifo_wr[active_slot[7]]) &
  fifo_llr_last[active_slot[7]]))
      begin
        if(InsertControlPointORType((slot_cnt[7]==(conf_slots_array[7]-4'd1)),
  control_test_point_en, tessent_persistent_cell_cp_test_in_1_0))
          slot_cnt[7] <= 3'd0;
        else
          slot_cnt[7] <= (slot_cnt[7] + 3'd1);
        end
      end
    end
  end

top1_rtl_tessent_dff_cp tessent_persistent_cell_cp_dff_instance_1( .clk(clk),
.ff_out(tessent_persistent_cell_cp_test_in_1_0) );

function InsertControlPointORType(input sig_in,  input TM, input test_in)
begin
      return (TM & test_in) | sig_in;
end
endfunction
```

Figure 3. Example RTL after test point insertion, showing maintained look and feel of the original RTL.

RTL test point analysis involves a comprehensive quick synthesis of the design and maps test point locations directly back to the RTL, where they are inserted. Tessent also evaluates the RTL to assess the design's suitability for test points. This suitability analysis categorizes the design as having high, medium or low suitability for RTL-based test points. This additional analysis helps designers streamline the design process, allowing them to focus on circuits that are most suitable for RTL test point insertion.

The test points are inserted directly into the RTL and must be maintained throughout synthesis and optimization. Tessent RTL Pro handles complex Verilog and SystemVerilog constructs while maintaining the look and feel of the original RTL when writing modified design files. Figure 3 shows how the look and feel of the original RTL is maintained once the test points have been mapped back to and inserted within the RTL.

Faster synthesis for productivity

An important aspect of shifting left in the design process is acquiring a better understanding of the design earlier in the flow. Synthesis refers to the process of transforming behavioral RTL code into a gate-level representation of the design. This is done in the presence of constraints and optimization settings which are applied to meet performance, power, and area (PPA) goals. One potential drawback of the synthesis process is the time needed to convert RTL into gates, map those gates to the defined process technology, and then apply all constraints and optimizations.

Tessent software includes quick synthesis, enabling designers to synthesize their design to basic (Tessent-internal) cells with no or very little optimization, without targeting PPA goals. This speeds up the turn-around-time of the synthesis process by an average of 40X. Designers are using this method today to speed up the learning about their designs. They can determine estimated pattern count and test coverage in a fraction of the time, when compared to running full synthesis with optimization targeting PPA goals. This allows outliers to be identified earlier in the design process.

Making the learning process faster allows DFT teams to feedback information about the coverage and testability of their designs to the RTL designers. Learnings may then be applied as RTL changes meant to improve coverage or reduce pattern count. For example, high estimated pattern count may result in a decision to insert test points directly into the RTL, thereby allowing both testability enhancement and the full synthesis process to account for those new cells during PPA optimization.

Conclusion

Tessent DFT solutions enable shift-left design flows, fully supporting designers in their efforts to reduce their turnaround time, to learn more about their circuits early in the design process and to find and fix problems earlier. With the introduction of Tessent RTL Pro, Tessent continues to provide the industry's most advanced DFT solutions for advanced design flows.

Because test points are an important part of meeting ISO 26262 standards for functional safety and automotive designs, inserting test points at RTL represents an improved solution for implementing VersaPoint and OST test points. This supports all downstream synthesis design flows. In addition, the ability to analyze and insert all major DFT logic, such as EDT, SSN and BIST logic, including wrapper cells and x-bounding logic, allows customers to extend shift-left initiatives by substantially enhancing the testability of their RTL designs.

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